

Wiring Overview

This document provides a complete overview of all electrical connections between the Arduino Nano and the peripheral components. Section 1 shows the consolidated pin mapping; Section 2 details each component individually.

Colour legend:

	Power supply (5 V / VIN)
	Common ground (GND)
	I ² C bus (SDA / SCL)
	Signal lines (PWM, analog, interrupt)

1 - Arduino Nano Pin Mapping

The table below lists every active pin on the Arduino Nano and its connection(s).

Arduino Nano Pin	Connected Component(s)	Signal / Function
VIN	9 V battery via power switch	Power in
5 V	Servo (red), MyoWare (red), MPU-6050 ×2 (VIN), DRV2605L (VIN), Potentiometer (pin 1)	Shared 5 V rail
3.3 V	MPU-6050 #2 - AD0 pin	I ² C address-select → 0x69
GND	Servo (brown), MyoWare (black), MPU-6050 ×2 (GND), DRV2605L (GND), Potentiometer (pin 3)	Common ground
D2	DRV2605L - INT	Haptic-done interrupt
D3 (PWM)	Servo (orange)	Servo command (PWM)
A0	Potentiometer - wiper (pin 2)	Gain / sensitivity (analog)
A1	MyoWare SIG (white)	EMG envelope (analog)
A3	Servo - feedback (white)	Servo angle feedback (analog)
A4 (SDA)	MPU-6050 (upper arm) + MPU-6050 (forearm) + DRV2605L	Shared I ² C data
A5 (SCL)	MPU-6050 (upper arm) + MPU-6050 (forearm) + DRV2605L	Shared I ² C clock

I²C bus note: both MPU-6050 IMUs and the DRV2605L haptic driver share the same SDA (A4) and SCL (A5) lines. The upper-arm IMU uses address 0x68 (AD0 low); the forearm IMU uses 0x69 (AD0 pulled to 3.3 V).

2 - Per-Component Wiring

Each subsection below details the individual pin connections for every peripheral.

Datan S1213 Servo

Component Pin	Connects To	Notes
Red (VCC)	Arduino 5 V	Power supply
Brown (GND)	Arduino GND	Common ground
Orange (SIG)	Arduino D3 (PWM)	PWM control signal
White (FB)	Arduino A3	Analog angle feedback

SparkFun MyoWare Muscle Sensor (AT-04-001)

Component Pin	Connects To	Notes
Red (VCC)	Arduino 5 V	Power supply
Black (GND)	Arduino GND	Common ground
White (SIG)	Arduino A1	EMG envelope output (analog)

MPU-6050 IMU - Upper Arm (Adafruit)

Component Pin	Connects To	Notes
VIN	Arduino 5 V	Power supply
GND	Arduino GND	Common ground
SDA	Arduino A4	I ² C data (shared bus)
SCL	Arduino A5	I ² C clock (shared bus)
AD0	Not connected	I ² C address = 0x68 (default)

MPU-6050 IMU - Forearm (Adafruit)

Component Pin	Connects To	Notes
VIN	Arduino 5 V	Power supply
GND	Arduino GND	Common ground
SDA	Arduino A4	I ² C data (shared bus)
SCL	Arduino A5	I ² C clock (shared bus)

Component Pin	Connects To	Notes
AD0	Arduino 3.3 V	Sets I ² C address = 0x69

DRV2605L Haptic Driver (Adafruit)

Component Pin	Connects To	Notes
VIN	Arduino 5 V	Power supply
GND	Arduino GND	Common ground
SDA	Arduino A4	I ² C data (shared bus)
SCL	Arduino A5	I ² C clock (shared bus)
INT	Arduino D2	Haptic-done interrupt

Drake haptic actuator

Component Pin	Connects To	Notes
Terminal 1	DRV2605L OUT+	Drives the haptic actuator
Terminal 2	DRV2605L OUT-	Drives the haptic actuator

Potentiometer

Component Pin	Connects To	Notes
Pin 1	Arduino 5 V	High reference
Pin 2 (wiper)	Arduino A0	Analog gain/sensitivity input
Pin 3	Arduino GND	Low reference (swap pins 1&3 to invert direction)